DATA ANALYST CASE STUDY

STARBUCKS

Global Retail Revenue & Customer Analytics

Company	Starbucks Corporation	**Industry**	QSR — Coffee & Retail
Tools	Excel | Power BI	**Difficulty**	Intermediate
Est. Time	8–10 hours		

1. Business Background

Starbucks Corporation is the world's largest coffeehouse chain, with 35,000+ stores across 80 markets. Its global retail arm processes millions of orders monthly across six country markets — Canada (CA), China (CN), Germany (DE), Japan (JP), United Kingdom (UK), and the United States (US) — through four payment channels (Cash, Credit Card, Mobile App, Starbucks Card) and three loyalty tiers (Non-Member, Starbucks Rewards, Starbucks Rewards Gold).

This case study uses three relational tables from the Starbucks retail operations system:

Table	Rows	Key Fields
Customers.csv	10,000	customer_id · country · membership_tier · signup_date
Orders.csv	10,000	order_id · customer_id · product_category · payment_method · order_status
Revenue.csv	9,652	order_id · amount_paid_usd · refund_amount_usd · payment_status

⚠ *Revenue.csv has 348 fewer rows than Orders.csv — a structural gap that is itself one of the four business problems.*

2. The Business Problem

Finance & Global Retail Operations runs on volume. One bad order tells you nothing — ten thousand orders tell you everything. Despite sitting on this data, the VP has no consolidated view across markets, payment types, or loyalty tiers. Four questions are open and unanswered:

3. Stakeholders

Stakeholder	Role	What They Need
VP of Finance & Global Retail Ops	Decision-maker	Country investment ranking · payment risk profile · loyalty programme ROI
Country Market Leads	Regional owners	Their market's performance vs. the other five on revenue, delivery rate, attrition
Loyalty Programme Manager	Tier strategy	Net revenue per customer by tier — evidence Gold justifies its operating cost
Payments & Fraud Team	Risk & compliance	Exact count of gap orders, their status (especially 'Delivered'), country/category breakdown
You — Data & Analytics	Analyst	Clean data, reproducible numbers, working dashboard, 1-page VP summary

4. Dataset Column Reference

Customers.csv

Column	Type	Meaning
customer_id	Text	Unique customer key — e.g. STA-C8580
country	Text	ISO-2 code: CA · CN · DE · JP · UK · US
membership_tier	Text	Non-Member | Starbucks Rewards | Starbucks Rewards Gold
signup_date	Date	Account creation date

Orders.csv

Column	Type	Meaning
order_id	Text	Unique order key — FK into Revenue.csv
product_category	Text	8 types: Frappuccino · Hot Coffee · Cold Coffee · Tea · Bakery · Sandwiches · Merchandise · Coffee Beans (Retail)
payment_method	Text	Cash | Credit Card | Mobile App | Starbucks Card
order_status	Text	Delivered | Shipped | Pending | Cancelled | Returned | Unknown

Revenue.csv

Column	Type	Meaning
order_id	Text	FK → Orders — 348 orders have NO matching row in this table
amount_paid_usd	Float	Gross amount collected before refunds ($0 – $431)
refund_amount_usd	Float	Refund issued — $0.00 for most rows, up to $194
payment_status	Text	Paid | Pending | Refunded

5. Data Quality Issues — Resolve Before Analysis

Each issue below, if ignored, produces wrong numbers silently. Clean the data first; build charts and measures second.

#	Issue	Why It Matters — What Goes Wrong If Ignored
1	Revenue has 9,652 rows for 10,000 orders — 348 orders have zero payment record.	Gap orders are invisible in SUM(amount_paid_usd). Revenue total is understated; order count is overstated. LEFT JOIN first; flag nulls before any metric is computed.
2	payment_status = 'Pending' on 643 revenue rows — money expected but not confirmed.	Including Pending inflates Net Revenue. A sale that has not cleared is not revenue. Exclude from all revenue KPIs; show separately as 'Revenue at Risk'.
3	order_status = 'Unknown' on 282 rows — a pipeline logging failure, not a real order state.	Unknown orders cannot be classified as Delivered, Cancelled, or Returned. Including them corrupts Delivery Rate and Attrition %. Isolate in a quarantine segment.
4	214 gap orders carry status = 'Delivered' — product shipped, no payment record exists.	Highest-severity issue: inventory left the store without confirmed payment. This is a financial exposure, not just a data defect. Escalate to Payments & Fraud separately from routine null handling.
5	refund_amount_usd is $0 for most rows, with a right-skewed tail to $194.	Averaging refunds is meaningless. Always SUM refunds. Refund Rate = SUM(refund) ÷ SUM(amount_paid). Never average on a per-row basis.
6	customer_id repeats across orders — order_id is the grain, not customer_id.	Deduplicating on customer_id collapses order history. Use COUNT(order_id) for volume and COUNT(DISTINCT customer_id) for reach. Never mix them in the same KPI.

⚠ Critical rule: Do not build any chart, pivot, or DAX measure until issues #1–#4 above are resolved. A dashboard built on uncleaned data gives the VP confident wrong answers — which is worse than no dashboard at all.

6. Your Deliverables

#	File	Must Contain
1	Starbucks.xlsx (Excel — Cleaning & EDA)	Cleaned data with documented decisions · LEFT JOIN flagging 348 gap orders · 4 labeled charts (Net Revenue by Country · Payment Risk Matrix · Loyalty Tier Revenue/Customer · Gap Heatmap) · 3+ written business insights
2	Starbucks.pbix (Power BI — Dashboard)	Star schema: Orders = fact, Customers + Revenue = lookup · All 9 DAX measures built before any visual · 5 dashboard pages with slicers by country, payment method, loyalty tier
3	VP Summary (1-page PDF or Word)	4 numbered findings, each backed by a specific number from your data · 3 concrete recommendations · 'Risk if ignored' statement for each

7. Business Questions the VP Needs Answered

You are not given the answers here. Your job is to find them. These are the four questions your cleaned data, your Excel analysis, and your Power BI dashboard must answer — with specific numbers, not vague observations:

The VP of Finance & Global Retail Operations needs answers on:

(1) Which of the 6 country markets deserve the next round of marketing/store investment,

(2) Whether Cash or Starbucks Card customers carry more payment-gap or refund risk than Credit Card or Mobile App customers,

(3) Which loyalty tier (Non-Member, Starbucks Rewards, Starbucks Rewards Gold) delivers the best net revenue, and

(4) How large the "order with no payment record" gap is and whether it concentrates in a particular country or category. Because Starbucks runs on volume rather than big-ticket sales, these questions can only be answered at scale — one bad order tells you nothing, ten thousand orders tell you everything.